Heritage Bank — Commercial Banking Team: Technology Context

Services Owned by This Team

Service	Responsibility	Technology
Commercial Account Service	Multi-account dashboard, balance aggregation, statement retrieval for business accounts	Spring Boot 3.2 → BIH → AS/400
ACH Batch Service	NACHA file upload, validation, dual-auth workflow, BIH submission, return handling	Spring Boot 3.2 + S3 + IBM MQ
Wire Service	Wire initiation, template management, Wire Desk review queue, Fedwire/SWIFT submission via BIH	Spring Boot 3.2 → BIH → AS/400
User Management Service	SMB user invitation, role assignment (Viewer/Initiator/Approver), per-account role model	Spring Boot 3.2 + PostgreSQL
Commercial Notification Service	ACH return alerts, wire status updates, low balance alerts for business accounts	Spring Boot 3.2 + SES + Twilio

Frontends Owned by This Team

Client	Technology	Notes
SMB Business Portal	React 18 SPA (TypeScript) + Apollo Client	Okta session-based auth; role-based UI rendering (Viewer/Initiator/Approver); hosted on CloudFront
Wire Desk Internal Tool	React 18 SPA (internal, mTLS)	Heritage Wire Desk staff; review queue, release and reject controls; Okta SAML auth

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Key Technical Constraints

Dual Authorization Enforcement

- The dual-auth rule (`initiatedBy ≠ approvedBy`) is enforced in the **ACH Batch Service and Wire Service** before submitting to BIH — not just in the UI
- User roles are stored in the User Management Service; each role grant is scoped to a specific account ID
- The API checks: (1) requester holds Approver role for this account, AND (2) `approvedBy ≠ initiatedBy` on the specific record
- Any attempt to approve one's own transaction returns HTTP 403 with a clear error message

NACHA File Processing

- NACHA files uploaded to S3 (pre-signed URL, max 50 MB)
- ACH Batch Service fetches from S3 and validates synchronously: record count, batch hash total, company ID format, effective date (must be a banking business day), routing number format per row

- Row-level errors returned as a structured JSON array: `[{ "row": 14, "field": "routingNumber", "error": "Invalid ABA routing number" }]`
- Validated files are stored in S3 (immutable, 7-year retention); metadata record written to PostgreSQL

AS/400 Batch Window (Commercial Impact)

- ACH files submitted after the cut-off but within the batch window are queued in IBM MQ; BIH processes them first thing when the window closes
- Wire initiation requests during batch window are queued; Wire Desk review can still proceed during the window; release to Fedwire/SWIFT is queued until after the window

Actimize Integration

- All commercial payment events (`ach.batch.submitted` , `wire.initiated` , `wire.released` , `transfer.initiated`) published to Kafka
- Actimize on-prem system has a Kafka consumer group subscribed to these topics
- Actimize alert payload returned via a separate REST webhook to the Commercial Notification Service; triggers immediate push + email to SMB Owner and Compliance Officer

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External Integrations (this team)

Integration	Protocol	Direction	Notes
BIH → AS/400	FiServ REST + IBM MQ	ACH/Wire Service → BIH → AS/400	ACH file submission, wire release, balance queries
FedACH (via BIH)	NACHA format via AS/400 ACH module	ACH Batch Service → BIH → FedACH	ACH origination and return processing
Fedwire / SWIFT (via BIH)	Fedwire XML / SWIFT MT103	Wire Service → BIH → Fedwire/SWIFT	Domestic and international wire transmission
NICE Actimize	Kafka (producer) + REST webhook (consumer)	Commercial services → Actimize; Actimize → Alert Service	Real-time transaction feed; fraud alert ingestion
Okta	OIDC + SAML	SMB Portal + Wire Desk Tool ← Okta	Auth; MFA; role-based access claims
Amazon S3	HTTPS (pre-signed URL)	SMB Portal → S3; ACH Batch Service ← S3	NACHA file upload and secure storage
Salesforce Financial Services Cloud	REST API	User Management Service → Salesforce	New user grants and relationship events synced to Salesforce CRM

Non-Functional Requirements (this team)

Category	Requirement

NACHA validation	Row-level error reporting; validation must complete within 10 seconds for files up to 10,000 rows
Dual-auth enforcement	Server-side check (not just UI); 403 response with error detail if violated
ACH return visibility	Returns appear in portal within 30 minutes of BIH processing the FedACH return file
Wire Desk SLA	Domestic wires: reviewed within 2 hours; international SWIFT: reviewed within 4 hours
Cut-off enforcement	Same-day ACH: 2:30 PM CT; next-day ACH: 8 PM CT; domestic Fedwire: 5 PM CT; SWIFT: 3 PM CT — enforced by the service, not AS/400
Audit logging	Every ACH batch and wire action (create, submit, approve, reject, release) logged to Audit Service with `userId`, `accountId`, `amount`, `timestamp`
Availability	99.9% for ACH Batch and Wire Services during business hours (7 AM–7 PM CT); planned maintenance in batch windows only